Description

Illustration





Figure caption

Solid hardboard panels. Images used under license from Shutterstock.com

The material

The class of wood product called hardboard is made by hot (140 - 200°C) compression of wood fibers from which the carbohydrates and other soluble constituents have been dissolved out, increasing the proportion of lignin, which acts as a binder. The resulting sheet products have one smooth, glossy face and one rough one. They are hard, stiff and, with the addition of a little urea-formaldehyde (UF) or phenol-formaldehyde (PF) binder, water resistant. Most have a density in the range 800 - 1040 kg/m3.
There are three basic types of hardboard are available - standard, medium and oil-tempered. Oil tempered boards are impregnated with oil making them stiffer, harder and more resistant to scratches and moisture. A number of specially hardboards are available for specific purposes. These have either embossed, perforated, plastic-faced or veneer surfaces. Most hardboards cannot be used outside because they absorb water.
The data in this record are for standard hardboard.

Compositional summary

Cellulose/Hemicellulose/Lignin/12%H2O

General properties

Density	800	-	1.05e3	kg/m^3
Price	* 0.166	-	0.264	GBP/kg
Date first used	-10000			

Material form that data applies to

Bulk	✓
Sheet	✓

Building system

Superstructure	✓

Enclosure	✓			
Interiors	✓			
Services	✗			

Mechanical properties

Young's modulus	*	2.3	-	5	GPa
Shear modulus	*	1	-	1.3	GPa
Bulk modulus	*	2.2	-	4.4	GPa
Bending modulus		2.5	-	5.5	GPa
Poisson's ratio	*	0.2	-	0.3	
Yield strength (elastic limit)	*	12	-	24	MPa
Tensile strength		20	-	40	MPa
Compressive strength		20	-	40	MPa
Bending strength		25	-	45	MPa
Elongation	*	0.44	-	0.54	% strain
Hardness - Vickers	*	9.8	-	12	HV
Fatigue strength at 10^7 cycles		9.2	-	11	MPa
Fracture toughness	*	1	-	2	MPa.m^0.5
Mechanical loss coefficient (tan delta)	*	0.011	-	0.017	

Thermal and combustion properties

Thermal conductor or insulator?		Good insulator			
Thermal resistivity		7.14	-	8.33	m.°C/W
Thermal expansion coefficient	*	10	-	15	µstrain/°C
Specific heat capacity		1.66e3	-	1.71e3	J/kg.°C
Maximum service temperature		120	-	140	°C
Minimum service temperature		-73	-	-23	°C
Flammability		Highly flammable			
Emissivity	*	0.81	-	0.9	

Hygro-thermal properties

Water absorption	*	12	-	35	%
Water vapor permeability		9.12e-13	-	2.01e-12	kg.m/s.m^2.Pa
Frost resistance		Good			

Electrical properties

Electrical conductor or insulator?		Poor insulator			
Electrical resistivity	*	1.05e13	-	2.45e15	µohm.cm
Dielectric constant (relative permittivity)	*	6	-	30	

Values marked * are estimates.
ANSYS, Inc. provides no warranty for this data.

Dissipation factor (dielectric loss tangent)	* 0.4	-	0.5	
Dielectric strength (dielectric breakdown)	* 1	-	2	MV/m

Optical properties

Natural color

Light brown

Transparency	Opaque	
Transmissivity	0	%

Critical Materials Risk

High critical material risk?	No

Acoustic properties

Sound absorption	Average
Sound isolation	Average

Processability (scale 1 = impractical to 5 = excellent)

Moldability	3	-	4
Machinability	5		

Durability

Water (fresh)	Limited use
Water (salt)	Limited use
Weak acids	Limited use
Strong acids	Unacceptable
Weak alkalis	Acceptable
Strong alkalis	Unacceptable
Organic solvents	Acceptable
UV radiation (sunlight)	Good
Wear resistance	Unacceptable
Industrial atmosphere	Limited use
Rural atmosphere	Limited use
Marine atmosphere	Limited use

Primary material production: energy and climate change

Climate change (CO_2-eq) (virgin grade production)	* 1.78	-	1.96	kg/kg
Embodied energy (virgin grade production)	* 35.6	-	39.2	MJ/kg
Water Usage	* 665	-	735	l/kg

Material processing: energy

Machining, coarse, embodied energy	* 16.1	-	18.6	MJ/kg
Machining, fine, embodied energy	* 19	-	23.5	MJ/kg
Machining, grinding, embodied energy	* 21.9	-	28.4	MJ/kg

Material processing: climate change

Machining, coarse, climate change (CO2-eq)	* 1.05	-	1.22	kg/kg
Machining, fine, climate change (CO2-eq)	* 1.24	-	1.54	kg/kg
Machining, grinding, climate change (CO2-eq)	* 1.43	-	1.86	kg/kg

Recycling and end of life

Recycle	✓			
Functional recycle	✗			
Climate change (CO2-eq), recycling	* 0.926	-	1.02	kg/kg
Embodied energy, recycling	* 29.5	-	32.5	MJ/kg
Recycle fraction in current supply	1			%
Combust for energy recovery	✓			
Heat of combustion (net)	* 19.7	-	20.7	MJ/kg
Combustion CO2	* 1.73	-	1.82	kg/kg
Landfill	✓			
Biodegrade	✓			
A renewable resource?	✓			

Supporting information

Typical uses

Wall and ceiling linings, paneling, partitions, joinery, kitchen units, furniture, chalk boards, caravan interiors, floor coverings, floor underlays, facing flush doors, shopfitting, display and exhibition work. Some eucalyptus based boards are suitable for floor surfaces subjected for light domestic traffic. Not recommended for exterior use or for use where subjected to direct wetting or high humidity conditions. However, some specific brands or proprietary products can be suitable.

Links

Reference